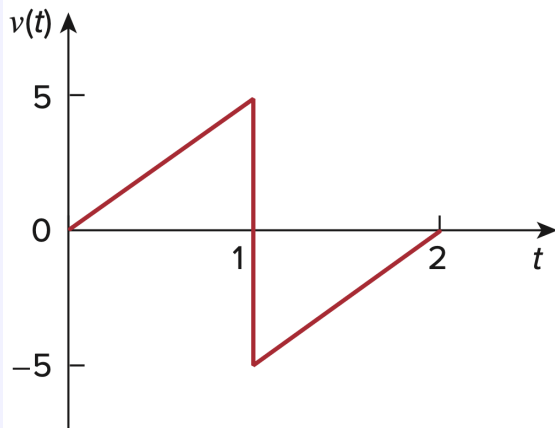


Exercise Sheet #5

Fields 2

Exercise 1: Time-Dependent Inductor Voltage and Current

If the voltage waveform below is applied to a 10-mH inductor, find the inductor current $i(t)$ for $0 < t < 2$ seconds. Assume $i(0) = 0$.



Exercise 2: Coil

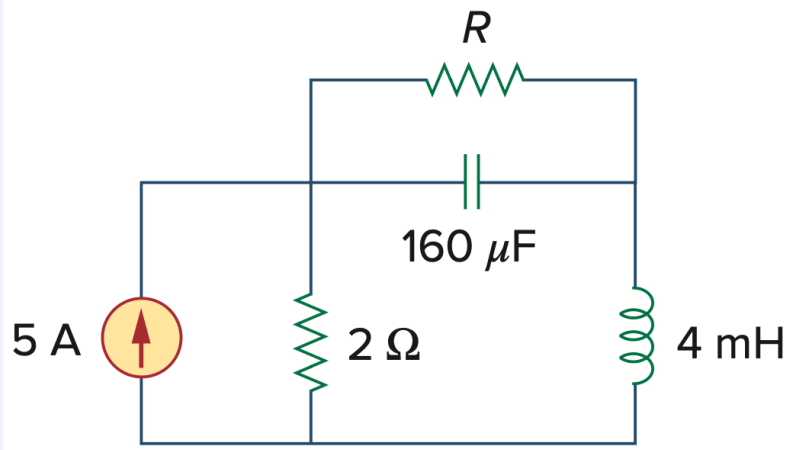
A cylindrical coil with a length of 2.7 m, a radius of 0.85 cm, and 600 turns carries a current of 2.5 A. Calculate B inside the coil, far from its ends.

Exercise 3: Energy of Inductors

The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

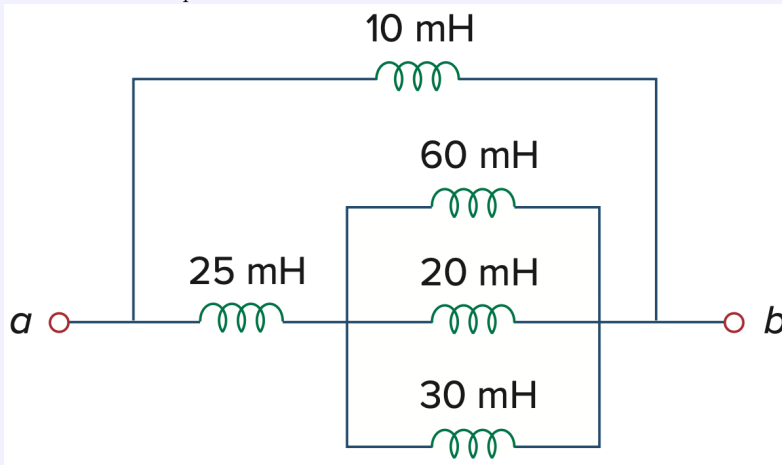
Exercise 4: Inductor and Capacitor under DC

For the circuit below, calculate the value of R that will make the energy stored in the capacitor the same as that stored in the inductor under DC conditions.



Exercise 5: Series and Parallel Inductors

Determine L_{eq} at terminals a-b of the circuit below.

**Exercise 6: Ideal Transformer**

A 480-V to 2400-V step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

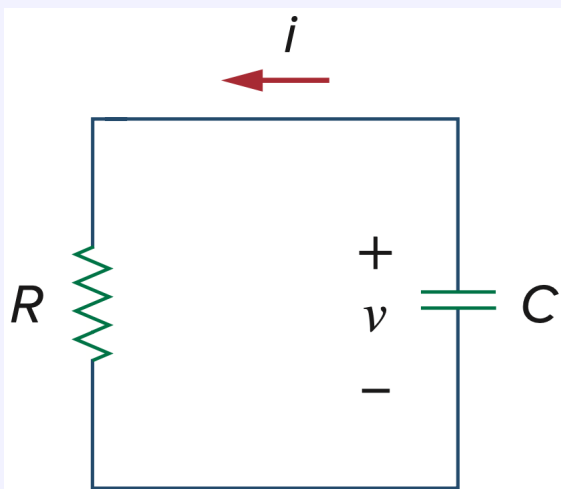
- the turns ratio
- the primary current
- the secondary current

Exercise 7: RC-Discharge

In the circuit shown below, for $t > 0$

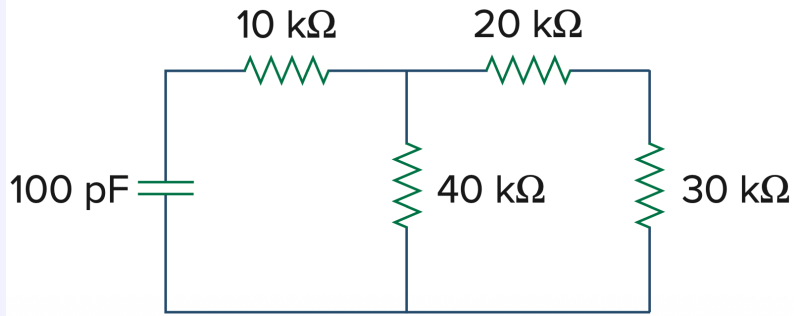
$$v(t) = 56e^{-200t} \text{ V}, \quad i(t) = 8e^{-200t} \text{ mA}.$$

- Find the values of R and C .
- Calculate the time constant τ .
- Determine the time required for the voltage to decay to half its initial value at $t = 0$.

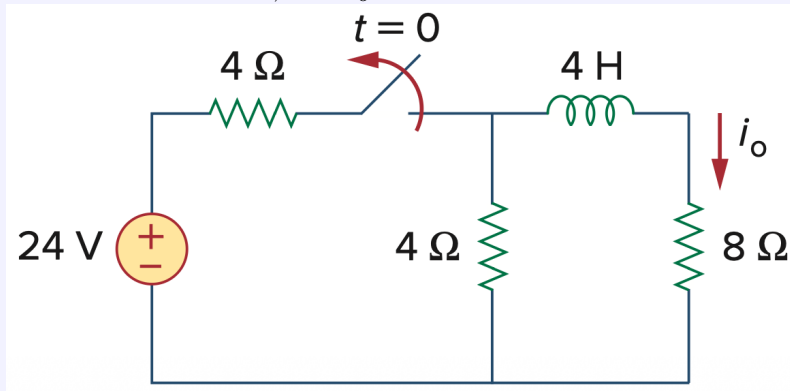


Exercise 8: Time Constant

Determine the time constant for the circuit below.

**Exercise 9: Inductor Current Decay**

For the circuit below, find i_o for $t > 0$.

**Exercise 10: Energy Dissipation during Transient**

In the circuit below,

$$v(t) = 80e^{-10^3 t} \text{ V}, \quad t > 0,$$

$$i(t) = 5e^{-10^3 t} \text{ mA}, \quad t > 0.$$

- Find R , L , τ .
- Calculate the energy dissipated in the resistance for $0 < t < 0.5 \text{ ms}$.

